

--	--	--	--	--	--	--	--	--	--

MCA
(SEM IV) THEORY EXAMINATION 2022-23
DATA ANALYTICS

Time: 3 Hours**Total Marks: 100****Note:** Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief. 2 x 10 = 20

- a. Explain various applications of data analytics.
- b. Discuss competitive learning.
- c. Explain stream computing.
- d. Explain CLIQUE.
- e. Name any four NoSQL databases.
- f. Explain R tool graphical user interface.
- g. Explain the method of mining frequent itemset.
- h. Explain the reason why Hadoop comes in market.
- i. Explain stochastic search method.
- j. What are key roles for successful analytic projects.

SECTION B

2. Attempt any three of the following: 10x3=30

- a. Explain various phases of data analytics life cycle.
- b. Explain rule induction with suitable example.
- c. Explain real time sentiment analysis.
- d. Discuss k-means clustering technique with suitable example.
- e. Explain HDFS architecture.

SECTION C

3. Attempt any one part of the following: 10x1=10

- a. Explain data, characteristics of data, and types of data.
- b. Explain the concept of Principal Component Analysis (PCM).

4. Attempt any *one* part of the following: 10x1=10

- a. Discuss Real -time analytics platform. Also, explain stock market prediction case study.
- b. Explain Apriori algorithm with example.

5. Attempt any *one* part of the following: 10x1=10

- a. Discuss NoSQL database and its types.
- b. Explain hierarchical clustering technique with example.

6. Attempt any *one* part of the following: 10x1=10

- a. Differentiate between
 - (1) HDFS and HBase
 - (2) HBase and HIVE
- b. Explain clustering in non-euclidean space.

7. Attempt any *one* part of the following: 10x1=10

- a. Differentiate between
 - (1) MapReduce and YARN
 - (2) Linear regression and Logistic regression
- b. Explain bayes theorem and concept learning.